DSAA3073: Theories in Data Science

Tutorial 1A: From Uncertainty to Geometry

175-minute core content master · delivery adaptation pending

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Explorative projection

LESSON BACKBONE

How can a finite sample and a geometric representation turn an uncertain data claim into a statement we can calculate and check?

Keep one learning problem in view. A predictor incurs a loss between 0 and 1 on each example. We observe only finitely many losses, yet we want to say something about the population mean. Probability will control the sampling error. Geometry will then give us a language for measuring parameters, correlations, and eventually functions.

The lesson has four core blocks: **tail control** (60 minutes), **norms and inner products** (55 minutes), **singular directions of a matrix** (25 minutes), and **functions as vectors** (35 minutes). The SVD block is a compact prerequisite for later spectral reasoning; numerical SVD algorithms and a proof of the full decomposition theorem remain outside this foundation. This is the content-master version, so the added block is recorded explicitly rather than hidden inside the earlier time budget.

Clock	Mathematical work	Protected time
0–60	Moment bounds and Hoeffding's inequality	60 min core
60–75	Scheduled break	15 min
75–130	Lp geometry and Cauchy–Schwarz	55 min core
130–155	Compact SVD and singular-direction geometry	25 min core
155–190	Functions as vectors and Hilbert-space structure	35 min core
After core	Questions, recovery, delivery adaptation	not part of the content master

From averages to guarantees

Suppose Z is the loss on a fresh example, with $0 \le Z \le 1$, and let $\mu = E[Z]$. From independent observations $Z_1, ..., Z_n$, we compute

$$\hat{\mu}_n = \frac{1}{n} \sum_{i=1}^{n} Z_i. \tag{1}$$

The number $\hat{\mu}_n$ is visible; μ is not. A statement such as "$\hat{\mu}_n$ is close to μ" is incomplete until we specify a tolerance and a probability. Our target is therefore

$$P(|\hat{\mu}_n - \mu| \ge \varepsilon) \le \delta. \tag{2}$$

Two questions are hidden in this line. Which assumptions justify the bound? How does the required sample size depend on ε and δ?

What can a mean alone buy?

Definition: Markov inequality

If X is a nonnegative random variable and $a > 0$, then $P(X \ge a) \le \frac{E[X]}{a}$.

The nonnegativity condition is doing real work. On the event $\{X \ge a\}$ we have $X \ge a$, so

$$E[X] \ge E\left[X \mathbf{1}_{\{X \ge a\}}\right] \ge aP(X \ge a). \tag{3}$$

Dividing by a gives the claim. Markov's inequality is deliberately modest: it uses only a mean and makes no claim about concentration around that mean.

Worked example: A useful bound with little information

Let $X \ge 0$ and $E[X] = 3$. Markov gives $P(X \ge 12) \le \frac{3}{12} = \frac{1}{4}$.

This is an upper bound, not the actual probability. It can be loose because every distribution with the same mean is treated alike.

To control deviation from a mean, apply Markov not to X, but to the nonnegative random variable $(X - E[X])^2$.

Definition: Chebyshev inequality

If X has finite mean μ and variance σ^2, then for every $\varepsilon > 0$,

$$P(|X - \mu| \ge \varepsilon) \le \frac{\sigma^2}{\varepsilon^2}. \tag{4}$$

Indeed, the events $\{|X - \mu| \ge \varepsilon\}$ and $\{(X - \mu)^2 \ge \varepsilon^2\}$ are identical. Markov then gives

$$P((X - \mu)^2 \ge \varepsilon^2) \le \frac{E[(X - \mu)^2]}{\varepsilon^2} = \frac{\sigma^2}{\varepsilon^2}. \tag{5}$$

For the sample mean of independent, identically distributed variables with variance σ^2, independence gives $\mathrm{Var}(\hat{\mu}_n) = \frac{\sigma^2}{n}$. Hence

$$P(|\hat{\mu}_n - \mu| \geq \varepsilon) \leq \frac{\sigma^2}{n\varepsilon^2}. \tag{6}$$

IN-CLASS CHECK

Prompt: A Bernoulli loss has variance at most $\frac{1}{4}$. Using only Chebyshev, how large must n be to make the deviation probability at most 0.05 when $\varepsilon = 0.1$?

Hint: Solve $\frac{1}{4n\varepsilon^2} \leq \delta$.

Answer:

> We need $n \geq \frac{1}{4\delta\varepsilon^2} = \frac{1}{4 \cdot 0.05 \cdot 0.01} = 500$. The calculation is valid for independent observations; without independence, the variance of the sample mean need not be $\frac{\sigma^2}{n}$.

Chebyshev improves as $\frac{1}{n}$. Bounded independent observations permit an exponential dependence on n instead.

Bounded independent averages

Theorem: Hoeffding inequality

Let $X_1, \ldots, X_n$ be independent random variables with $a_i \leq X_i \leq b_i$ almost surely. Then, for every $t > 0$,

$$P\left(\sum_{i=1}^{n}(X_i - E[X_i]) \geq t\right) \leq \exp\left(-2\frac{t^2}{\sum_{i=1}^{n}(b_i - a_i)^2}\right). \tag{7}$$

Applying the same result to the negative deviation gives a two-sided bound.

The proof skeleton shows exactly what the two assumptions buy. Write $S = \sum_i (X_i - E[X_i])$ and choose $\lambda > 0$. Exponential Markov gives

$$P(S \geq t) \leq e^{-\lambda t} E\left[e^{\lambda S}\right]. \tag{8}$$

For a random variable Y, its **moment-generating function** is $M_Y(\lambda) = E\left[e^{\lambda Y}\right]$ whenever this expectation is finite. Now **independence** is used once, to factor that function for the centered sum:

$$E\left[e^{\lambda S}\right] = \prod_{i=1}^{n} E\left[e^{\lambda (X_i - E[X_i])}\right]. \tag{9}$$

For a variable in $[a_i, b_i]$, **boundedness** gives Hoeffding's lemma,

$$E\left[e^{\lambda (X_i - E[X_i])}\right] \leq \exp\left(\lambda^2 \frac{(b_i - a_i)^2}{8}\right). \tag{10}$$

Combining the three lines yields $\exp\Big(-\lambda t + \lambda^2 \sum_i \frac{(b_i-a_i)^2}{8}\Big)$. Choosing $\lambda = 4\frac{t}{\sum_i (b_i-a_i)^2}$ produces the stated exponent. Thus independence creates the product, while boundedness controls each factor; neither assumption is decorative.

For independent losses $Z_i \in [0,1]$, set $t = n\varepsilon$. The two tails give

$$P(|\hat{\mu}_n - \mu| \geq \varepsilon) \leq 2\exp(-2n\varepsilon^2). \tag{11}$$

Read the conclusion together with its assumptions:

- the Z_i are independent;
- every Z_i is bounded in a known interval;
- $\varepsilon > 0$ is an absolute deviation in the same units as the loss.

The inequality is distribution-free within those conditions. It does not say the loss is Gaussian, and it does not need the variance to be known.

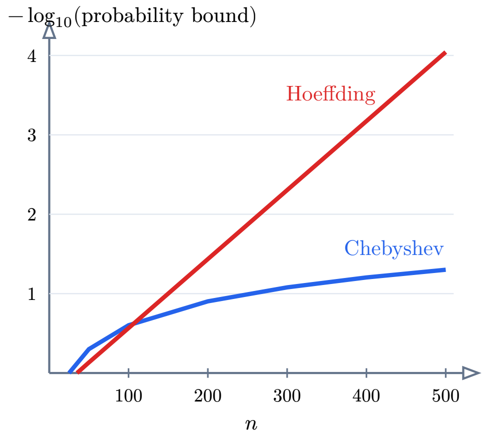
Solving the guarantee backward

To ensure $2\exp(-2n\varepsilon^2) \leq \delta$, take logarithms:

$$-2n\varepsilon^2 \leq \ln\left(\frac{\delta}{2}\right) \quad \rightarrow \quad n \geq \frac{\ln\left(\frac{2}{\delta}\right)}{2\varepsilon^2}. \tag{12}$$

> **Worked example: How many bounded losses?**
>
> With $\varepsilon = 0.1$ and $\delta = 0.05$,
>
> $$n \geq \frac{\ln(40)}{0.02} \approx 184.44. \tag{13}$$
>
> Thus $n = 185$ is sufficient under Hoeffding. Chebyshev required 500 under the variance-only calculation above. Neither number is an exact minimum; each is sufficient according to its bound.

The plot holds $\varepsilon = 0.1$ fixed and changes only n. Its vertical scale is $-\log_{10}(\text{bound})$, so moving upward means a smaller probability bound. For Bernoulli losses, Chebyshev uses $\min(1, \frac{25}{n})$ and Hoeffding uses $\min(1, 2e^{-0.02n})$.



Curve	Assumptions used
Chebyshev	finite variance; independence for $\mathrm{Var}(\hat{\mu}_n) = \frac{\sigma^2}{n}$
Hoeffding	independent observations bounded in $[0, 1]$

Table 1: Polynomial and exponential upper bounds for the same deviation event on a logarithmic probability scale.

IN-CLASS CHECK

Prompt: A dataset is collected as a time series with strong dependence. May we insert its observations into the displayed Hoeffding bound merely because every loss lies in $[0, 1]$?

Hint: List every assumption before looking at the formula.

Answer:

> No. Boundedness holds, but independence has not been justified. The correct response is not to use the formula with a warning label; it is to establish an appropriate dependence-aware result or change the sampling argument.

IN-CLASS CHECK

Prompt: Before calculating, predict which curve moves farther upward when n doubles from 100 to 200. Then verify both bounds.

Hint: Chebyshev is proportional to $\frac{1}{n}$; Hoeffding is exponential in n.

Answer:

> Chebyshev changes from 0.25 to 0.125, so it halves. Hoeffding changes from $2e^{-2} \approx 0.271$ to $2e^{-4} \approx 0.0366$. The Hoeffding curve therefore moves much farther upward on the logarithmic scale. Algebraically, if $B(n) = 2e^{-2n\varepsilon^2}$, then $B(2n) = \frac{B(n)^2}{2}$.

Key Takeaway

A tail bound is a contract between assumptions and a conclusion. A mean supports Markov; a variance supports Chebyshev; bounded independent summands support Hoeffding. Stronger conclusions do not come from algebra alone.

The geometry hidden in a norm

Our learning claim now has probabilistic control, but a predictor is often represented by a vector of parameters. To discuss the size of that vector, the effect of a perturbation, or the alignment between two directions, we need a geometric language.

One vector, several notions of size

Definition: Lp norm

For $x = (x_1, ..., x_d) \in \mathbb{R}^d$ and $1 \leq p < \infty$,

$$\|x\|_p = \left(\sum_{j=1}^{d} |x_j|^p \right)^{\frac{1}{p}}. \tag{14}$$

The limiting case is $\|x\|_\infty = \max_{1 \leq j \leq d} |x_j|$. The subscript p is part of the meaning and must be stated when the geometry matters.

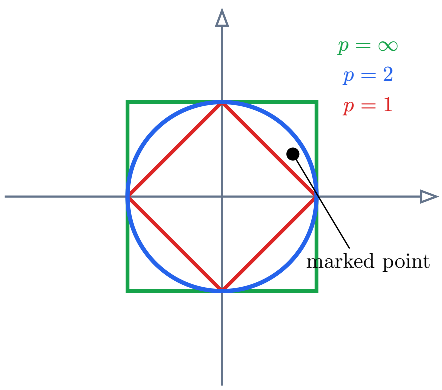
This is the course-wide notation: $\|x\|_p$. The same vector can have different numerical sizes because p changes how coordinates are aggregated.

Worked example: The vector $(3, -4)$

$$\|(3, -4)\|_1 = 7, \quad \|(3, -4)\|_2 = 5, \quad \|(3, -4)\|_\infty = 4. \tag{15}$$

None is "the true length" without a declared geometry. The Euclidean norm $\|x\|_2$ is the usual geometric length; $\|x\|_1$ accumulates coordinate magnitudes; $\|x\|_\infty$ records the largest coordinate magnitude.

The three unit balls below share the same axes and scale. Only the value of p changes.



IN-CLASS CHECK

Prompt: For the marked point $x = (0.75, 0.45)$, decide whether it is inside each unit ball.

Hint: Compute $\|x\|_1$, $\|x\|_2$, and $\|x\|_\infty$.

Answer:

> $\|x\|_1 = 1.20$, so the point is outside the L1 ball. $\|x\|_2 = \sqrt{0.75^2 + 0.45^2} \approx 0.875$, so it is inside the L2 ball. $\|x\|_\infty = 0.75$, so it is inside the Linfinity ball.

In a fixed finite dimension, these norms control one another. If $1 \le p \le q \le \infty$, then

$$\|x\|_q \le \|x\|_p \le d^{\frac{1}{p}-\frac{1}{q}}\|x\|_q. \tag{16}$$

Consequently, convergence under one Lp norm is equivalent to convergence under another when d is fixed. **Equivalent does not mean interchangeable:** the dimension-dependent constants and the shapes of the level sets still change an optimization problem.

The formula $\left(\sum_j |x_j|^p\right)^{\frac{1}{p}}$ is a norm only for $p \ge 1$. For $0 < p < 1$ it can violate the triangle inequality. We will not call that quantity a norm.

Inner products control alignment

For $x, y \in \mathbb{R}^d$, the Euclidean inner product is $\langle x, y\rangle = \sum_{j=1}^d x_j y_j$. This is the course-wide inner-product notation for both finite-dimensional and Hilbert spaces.

Theorem: Cauchy–Schwarz inequality

For vectors $x, y \in \mathbb{R}^d$ with the Euclidean inner product and norm,

$$|\langle x, y\rangle| \le \|x\|_2\|y\|_2. \tag{17}$$

Equality holds exactly when x and y are linearly dependent, including the zero-vector cases.

A useful proof turns the geometric claim into a one-variable nonnegativity argument. If $y = 0$, the claim is immediate. For $y \ne 0$, every real t satisfies

$$0 \le \|x - ty\|_2^2 = \|x\|_2^2 - 2t\langle x, y\rangle + t^2\|y\|_2^2. \tag{18}$$

Choose $t = \frac{\langle x,y\rangle}{\|y\|_2^2}$. Substitution gives

$$0 \le \|x\|_2^2 - \frac{(\langle x, y\rangle)^2}{\|y\|_2^2}, \tag{19}$$

which rearranges to the squared inequality. Equality means $x - ty = 0$, so the two vectors are linearly dependent.

Worked example: A strict case

Let $x = (1, 2, -1)$ and $y = (2, 0, 3)$. Then

$$\langle x, y\rangle = 2 + 0 - 3 = -1, \quad \|x\|_2 = \sqrt{6}, \quad \|y\|_2 = \sqrt{13}. \tag{20}$$

Thus $|\langle x, y\rangle| = 1 < \sqrt{78}$. The inequality is strict because x and y are not scalar multiples.

IN-CLASS CHECK

Prompt: Find a nonzero vector y for which equality holds with $x = (1, -2, 3)$, and verify the equality numerically.

Hint: Choose any scalar multiple of x.

Answer:

Take $y = 2x = (2, -4, 6)$. Then $\langle x, y\rangle = 28$, $\|x\|_2 = \sqrt{14}$, and $\|y\|_2 = 2\sqrt{14}$, so the product of norms is 28. Both sides agree.

Cauchy–Schwarz is not merely a fact about angles. It converts a difficult inner product into a product of sizes. That pattern reappears whenever a proof needs to separate two interacting quantities.

IN-CLASS CHECK

Prompt: Maximize the score $2x_1 + x_2$ first under $\|x\|_1 \le 1$ and then under $\|x\|_2 \le 1$. Which maximizing vector is sparse, and which is diffuse?

Hint: For the L1 ball, compare its vertices. For the L2 ball, use Cauchy–Schwarz with $(2, 1)$.

Answer:

On the L1 ball the maximum is attained at $x = (1, 0)$, with score 2; this vector is sparse. On the L2 ball, Cauchy–Schwarz gives a maximum of $\sqrt{5}$ at $x = \frac{2,1}{\sqrt{5}}$; both coordinates are nonzero, so the solution is diffuse. An Lp penalty has these same level-set shapes. This is why changing a norm can change solution structure even though all norms are equivalent in fixed dimension.

Key Takeaway

A norm measures size after a geometry has been chosen. An inner product also measures alignment, and Cauchy–Schwarz limits that alignment by the two induced sizes. The symbols $\|x\|_p$ and $\langle x, y\rangle$ will keep these roles throughout the course.

A matrix through its singular directions

A norm measures one vector, but a matrix acts on every direction. Looking only at its entries can hide three different behaviors: some input directions are stretched strongly, some weakly, and some are erased. We need a representation that exposes those directions without assuming that the matrix is square or invertible.

Begin with $A = \operatorname{diag}(4, \frac{1}{2}, 0)$. The coordinate directions are already special: e_1 is stretched by 4, e_2 by $\frac{1}{2}$, and e_3 is sent to zero. The singular-value decomposition generalizes this direction-by-direction description to an arbitrary real matrix.

The compact singular-value decomposition

Definition: Compact singular-value decomposition

Let $A \in \mathbb{R}^{m\times d}$ have rank r. Its **compact singular-value decomposition** is

$$A = U_r \Sigma_r V_r^T, \tag{21}$$

where $U_r \in \mathbb{R}^{m\times r}$ and $V_r \in \mathbb{R}^{d\times r}$ have orthonormal columns, and

$$\Sigma_r = \operatorname{diag}(\sigma_1, ..., \sigma_r), \quad \sigma_1 \geq ... \geq \sigma_r > 0. \tag{22}$$

The columns u_j of U_r are the left singular vectors, the columns v_j of V_r are the right singular vectors, and the positive numbers σ_j are the singular values.

The registered course form is $A = U_r \Sigma_r V_r^T$. Its dimensions make the input-output roles visible:

$$\underbrace{U_r}_{m\times r} \underbrace{\Sigma_r}_{r\times r} \underbrace{V_r^T}_{r\times d} \quad = \quad \underbrace{A}_{m\times d}. \tag{23}$$

Multiplying by a right singular vector gives the central relation

$$Av_j = \sigma_j u_j, \quad A^T u_j = \sigma_j v_j. \tag{24}$$

Thus v_j is an orthogonal input direction, u_j is the matching output direction, and σ_j is the stretch factor. The compact form keeps only the positive singular pairs. Zero singular directions are recorded separately through the kernel.

For $A = \operatorname{diag}(4, \frac{1}{2}, 0)$, the direction ledger is

Input direction	Stretch	Output	Inverse action
e_1	4	$4e_1$	divide by 4
e_2	$\frac{1}{2}$	$(\frac{1}{2})e_2$	divide by $\frac{1}{2}$
e_3	0	0	not recoverable from the output

The compact SVD uses $r = 2$,

$$U_r = V_r = [e_1 \quad e_2], \quad \Sigma_r = \operatorname{diag}\left(4, \frac{1}{2}\right). \tag{25}$$

The table is not special to diagonal matrices: an arbitrary SVD first rotates to the orthogonal directions in V_r, scales them by Σ_r, and expresses the result in the orthogonal output directions in U_r.

Rank, range, kernel, and inverse amplification

The compact SVD turns several linear-algebra objects into readable subspaces:

$$\operatorname{range}(A) = \operatorname{span}\{u_1, ..., u_r\}, \quad \operatorname{range}(A^T) = \operatorname{span}\{v_1, ..., v_r\}, \tag{26}$$

$$\ker(A) = \operatorname{span}\{v_1, ..., v_r\}^{\perp}, \quad \operatorname{rank}(A) = r. \tag{27}$$

On the active directions, the **Moore–Penrose pseudoinverse** reverses each positive stretch:

$$A^{\dagger} = V_r \Sigma_r^{-1} U_r^T, \quad A^{\dagger} u_j = \frac{1}{\sigma_j} v_j. \tag{28}$$

It does not reconstruct a direction in $\ker(A)$; the data contain no information about that component. The induced Euclidean operator norm is

$$\|A\|_2 = \sup_{x \neq 0} \frac{\|Ax\|_2}{\|x\|_2} = \sigma_1. \tag{29}$$

A small positive singular value has a direct qualitative consequence: the pseudoinverse multiplies the corresponding output component by $\frac{1}{\sigma_j}$. Thus a small perturbation can be strongly amplified when we try to recover an input. A zero singular direction is more severe: that component is not recoverable at all. Chapter 4 will formalize numerical condition numbers; here we keep only the singular-direction mechanism that makes instability visible.

IN-CLASS CHECK

Prompt: For $A = \operatorname{diag}(3, 0.1, 0)$, state its rank, range, kernel, Euclidean operator norm, compact pseudoinverse, and the amplification of a perturbation in the second active output direction.

Hint: Read every answer from the singular directions; invert only positive singular values.

Answer:

> The rank is 2, the range is $\operatorname{span}\{e_1, e_2\}$, and the kernel is $\operatorname{span}\{e_3\}$. The operator norm is 3. The pseudoinverse is $A^{\dagger} = \operatorname{diag}(\frac{1}{3}, 10, 0)$. A perturbation in the e_2 output direction is multiplied by $\frac{1}{0.1} = 10$ when mapped back to the active input direction. The zero direction remains unrecoverable.

Key Takeaway

SVD is a direction-by-direction description of a linear map. The right singular vectors are input directions, the left singular vectors are output directions, and the singular values are stretches. Rank counts the positive pairs; the kernel contains erased directions; the pseudoinverse divides only by positive singular values. Later inverse-problem calculations may use this compact SVD language directly.

When the vectors are functions

The algebra of an inner product does not require a finite coordinate list. To make the function version precise, work in $L^2([0,1])$: the real functions with

$$\int_0^1 |f(t)|^2 \,\mathrm{d}t < \infty. \tag{30}$$

Functions that differ only on a set of measure zero are treated as the same element. This is the meaning of equality **almost everywhere**: equality may fail on a set whose total length is zero. For elements of $L^2([0,1])$, define

$$\langle f, g\rangle = \int_0^1 f(t)g(t)\,\mathrm{d}t, \quad \|f\|_2 = \sqrt{\int_0^1 |f(t)|^2\,\mathrm{d}t}. \tag{31}$$

Addition and scalar multiplication are pointwise (up to almost-everywhere equality). The inner product returns a number, exactly as a dot product does.

Definition: Hilbert space

An inner-product space H is a Hilbert space when it is complete in the induced norm $\|h\| = \sqrt{\langle h, h\rangle}$. A sequence (h_n) is Cauchy in this norm if, for every $\eta > 0$, there is an N such that $\|h_m - h_n\| < \eta$ whenever $m, n \geq N$. Completeness means every such sequence converges in norm to an element of H.

The space $L^2([0,1])$ is a Hilbert space. The word **complete** prevents a limiting process from leaving the space. We do not need the proof of completeness here; we need the structural promise that inner products, norms, limits, and Cauchy–Schwarz coexist where a function-valued optimization problem is posed.

Operation	**Finite vector**	**Function on $[0,1]$**
Object	$(x_1, \dots, x_d)$	$f(t)$, shown by its curve over t
Addition	$x+y$	$(f+g)(t) = f(t) + g(t)$
Scaling	cx	$(cf)(t) = cf(t)$
Inner product	$\sum_j x_j y_j$	$\int_0^1 f(t)g(t)\,\mathrm{d}t$
Induced size	$\sqrt{\langle x, x\rangle}$	$\sqrt{\langle f, f\rangle}$
Example shapes	$(1,1)$ and $(1,-1)$	$f(t) = 1$ and $g(t) = 2t-1$

$f(t) = 1$ **$g(t) = 2t - 1$**

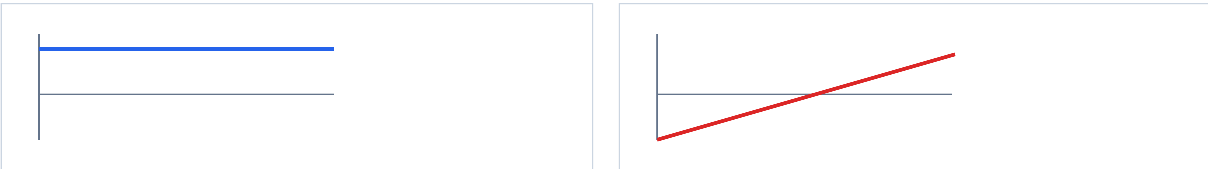


Table 2: The same algebraic questions in a coordinate space and a function space. Both plots use the same vertical scale: $g(t) = 2t - 1$ starts below zero, crosses at $t = \frac{1}{2}$, and ends above zero.

Worked example: Two orthogonal functions

Let $f(t) = 1$ and $g(t) = 2t - 1$ on $[0,1]$. Then

$$\langle f, g\rangle = \int_0^1 (2t-1)\,\mathrm{d}t = \left[t^2 - t\right]_0^1 = 0. \tag{32}$$

Their norms are

$$\|f\|_2 = 1, \quad \|g\|_2 = \sqrt{\int_0^1 (2t-1)^2\,\mathrm{d}t} = \frac{1}{\sqrt{3}}. \tag{33}$$

The functions are nonzero but orthogonal, just as perpendicular nonzero vectors have zero dot product.

Cauchy–Schwarz extends to every inner-product space once its induced norm has been defined. In $L^2([0,1])$ it reads

$$\left|\int_0^1 f(t)g(t)\,\mathrm{d}t\right| \le \sqrt{\int_0^1 |f(t)|^2\,\mathrm{d}t}\sqrt{\int_0^1 |g(t)|^2\,\mathrm{d}t}. \tag{34}$$

IN-CLASS CHECK

Prompt: Let $f(t) = t$ and $g(t) = 1$ on $[0, 1]$. Compute the inner product and both norms, then verify Cauchy–Schwarz.

Hint: Use $\int_0^1 t\,\mathrm{d}t = \frac{1}{2}$ and $\int_0^1 t^2\,\mathrm{d}t = \frac{1}{3}$.

Answer:

> $\langle f, g\rangle = \frac{1}{2}$, $\|f\|_2 = \frac{1}{\sqrt{3}}$, and $\|g\|_2 = 1$. Since $\frac{1}{2} \le \frac{1}{\sqrt{3}}$, the inequality holds. It is strict because t is not a constant multiple of 1 almost everywhere.

IN-CLASS CHECK

Prompt: A space has the integral inner product above, and every norm-Cauchy sequence has a norm limit that remains in the space. Which two ingredients certify that it is a Hilbert space?

Hint: Reconstruct the definition rather than naming examples.

Answer:

> First, the displayed integral must define an inner product and hence an induced norm. Second, the space must be complete in that norm: each Cauchy sequence converges to an element still in the space. Inner-product structure plus completeness is exactly the Hilbert-space condition.

A later chapter will estimate functions while controlling a Hilbert-space norm as a penalty. No regularization machinery is needed today; the point of completeness is that norm-controlled limiting arguments stay inside the chosen function space.

What Tutorial B may assume

Tutorial B begins from exactly three pieces of the handoff below.

1. **Hoeffding inequality.** For independent $Z_i \in [0, 1]$, $P(|\hat{\mu}_n - \mu| \ge \varepsilon) \le 2\exp(-2n\varepsilon^2)$. You should be able to solve this bound for n or ε and state the independence and boundedness conditions.
2. **Lp norm.** The canonical form is $\|x\|_p$; the value of p declares the geometry. You should be able to compute the $p = 1, 2, \infty$ cases for a finite vector.
3. **The notation $\|x\|_p$.** The symbol and its subscript retain the meaning just stated; Tutorial B will not rename it.

Cauchy–Schwarz, compact SVD, the inner-product notation, and Hilbert-space structure remain part of the Chapter 1 foundation. Tutorial B does not require them, but Chapter 4 may assume that you can reconstruct $A = U_r \Sigma_r V_r^T$, $Av_j = \sigma_j u_j$, and the roles of the range, kernel, pseudoinverse, operator norm, and positive singular values.

Standard language to carry forward

Term	Meaning here	Representative form
Markov inequality	A nonnegative variable's mean controls one upper tail.	$P(X \geq a) \leq \frac{E[X]}{a}$
Chebyshev inequality	Variance controls deviation from the mean.	$P(\lvert X-\mu\rvert \geq \varepsilon) \leq \frac{\sigma^2}{\varepsilon^2}$
Hoeffding inequality	Bounded independent summands have exponential tail control.	$2e^{-2n\varepsilon^2}$ for $[0,1]$ means
Lp norm	A value of p selects how coordinate magnitudes determine size.	$\lVert x\rVert_p$
Cauchy–Schwarz inequality	Inner-product magnitude is controlled by induced sizes.	$\lvert\langle x,y\rangle\rvert \leq \lVert x\rVert_2\lVert y\rVert_2$
Compact SVD	A rank-r matrix is resolved into orthogonal input/output directions and positive stretches.	$A = U_r\Sigma_r V_r^T$
Pseudoinverse	Invert each positive singular value and leave kernel directions unrecovered.	$A^\dagger = V_r\Sigma_r^{-1}U_r^T$
Hilbert space	A complete inner-product space.	$\lVert f\rVert = \sqrt{\langle f,f\rangle}$

Key Takeaway

Probability told us when a sample average can stand in for an expectation. Geometry told us how to measure a vector, control an interaction, and read a matrix one singular direction at a time before extending the same language to functions. Tutorial B asks a different question: when does the shape and curvature of an objective make a useful computation possible?

Source notes

- M. Mohri, A. Rostamizadeh, and A. Talwalkar, *Foundations of Machine Learning*, 2nd ed.: Appendix A.1 (norms and Cauchy-Schwarz), Appendix A.2.1-A.2.2 (operator norm, singular-value decomposition, and pseudoinverse), Appendices C.4–C.5 (Markov and Chebyshev), and Appendix D.1 (Hoeffding). The tutorial writes the pseudoinverse as $A^\dagger = V_r\Sigma_r^{-1}U_r^T$, the dimensionally compatible order, rather than reproducing the reversed factor order printed in the Appendix A.2.2 display.
- H. Daumé III, "From Zero to Reproducing Kernel Hilbert Spaces in Twelve Pages or Less," Sections 4–6 (inner-product spaces, norms, completeness, and Hilbert spaces).
- DSAA3073 Week 1 Learning Sheet v4, pp. 1–22, used for course scope and sequencing; the original sources above control the mathematical statements.